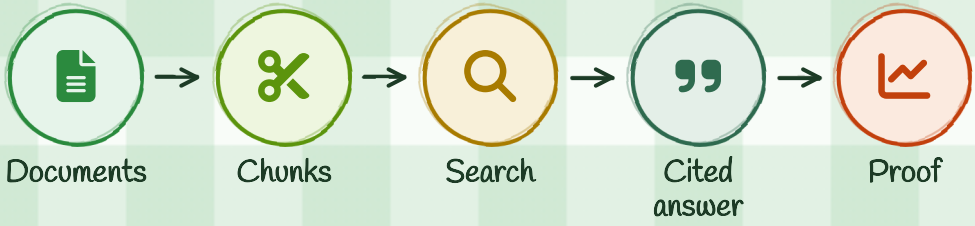
RAG ROADMAP 2026

The RAG Roadmap Nobody Draws Fully

Embed, retrieve, generate is about a third of the job

By **Naresh Edagotti** · PracticAI



14 steps, in the order you'd actually build them.

Swipe »

Why most demos fail

The Quick Prototype Breaks

Embed it all, take the top 3, ask

In the course story, six support agents tried exactly that for two weeks. It:

Stale
Served archived warranty terms as if they were current

Leaky
Showed finance-only cost prices to support agents

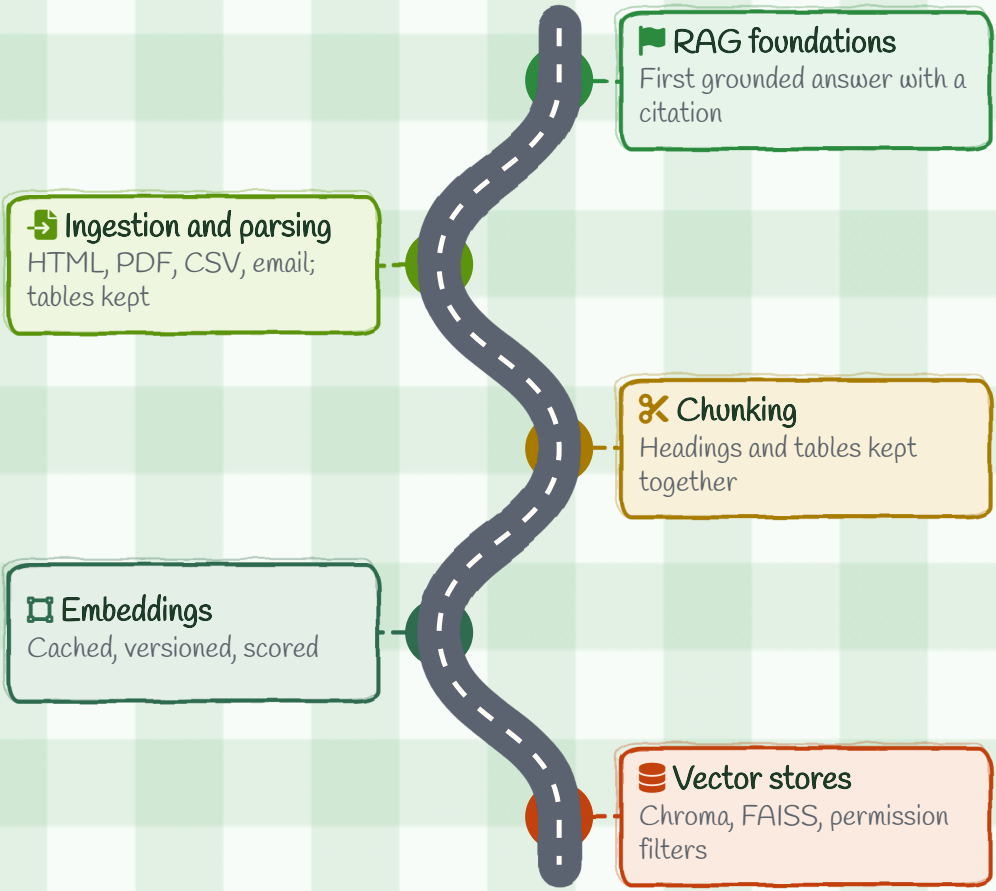
Hijacked
Ranked a forum post with an injected instruction first

Overconfident
Answered a question the documents could not answer

Every step on the roadmap **fixes one of these.**

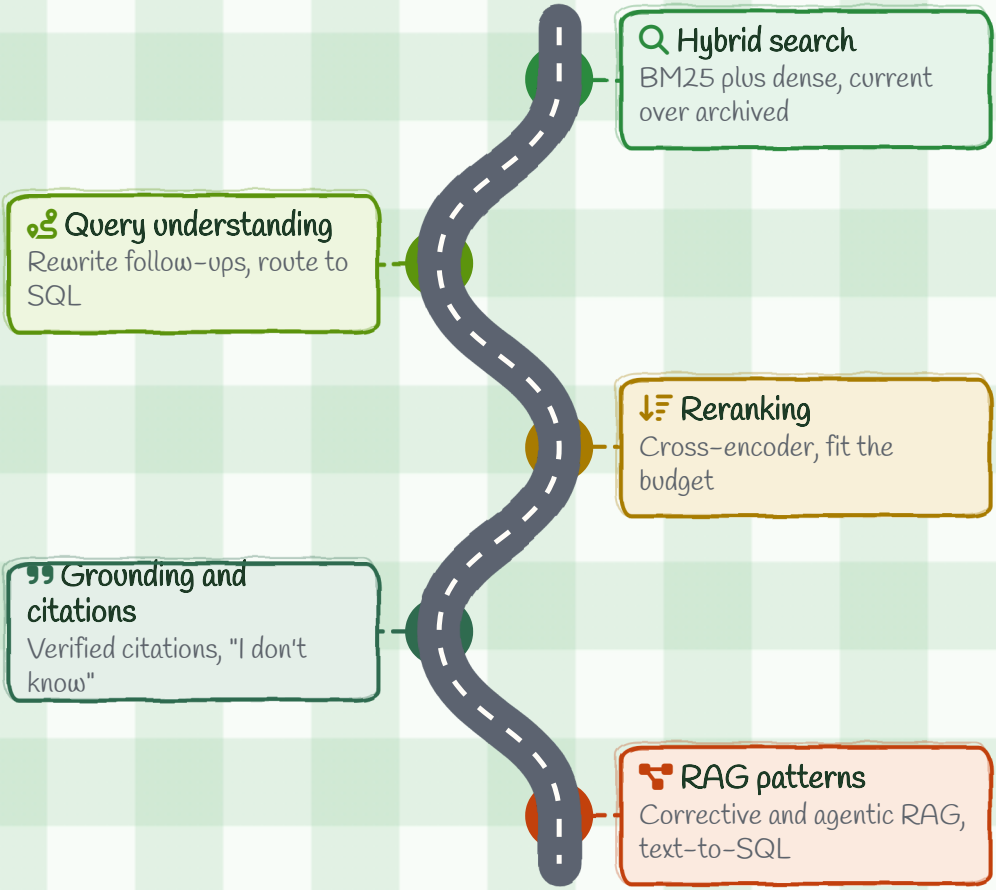
The roadmap, part 1

Steps 1 to 5: Documents In



The roadmap, part 2

Steps 6 to 10: The Right Piece

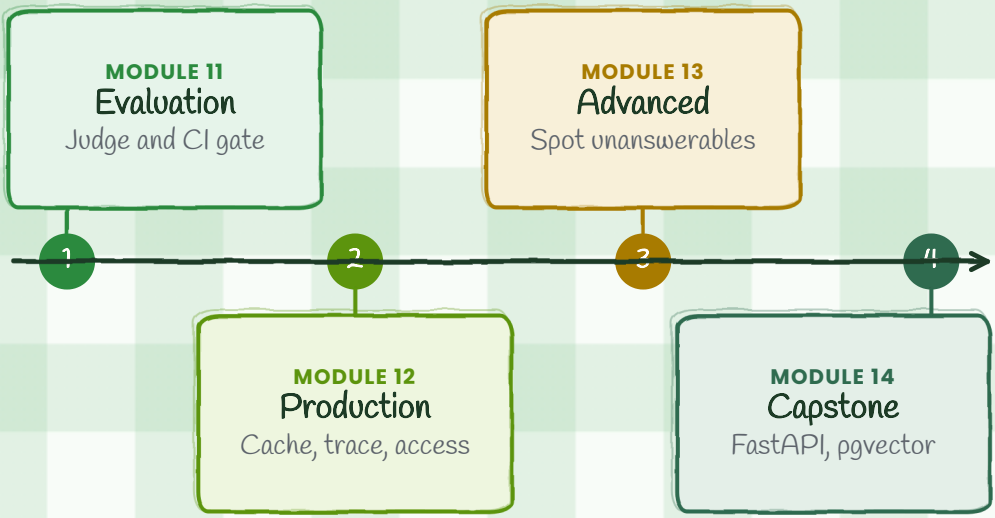


The roadmap, part 3

Steps 11 to 14: Proving It Works

The part most tutorials leave out

Measure it, harden it, then ship it as a real service.



If you can't measure it, you **can't ship it.**

Mindset shift

Demo RAG vs Production RAG

× Demo RAG

- × Fixed-size chunks, tables cut
- × Top 3 by similarity
- × Every user sees every doc
- × Answers even with no source
- × Looks right in a notebook

✓ Production RAG

- ✓ Chunks keep tables together
- ✓ Hybrid search, then rerank
- ✓ Permission filter in the search
- ✓ Says "I don't know"
- ✓ Proven by a regression gate

Same idea. **A very different system.**

The pipeline you build

Every Layer, Every Tool

Ingest	PDF · Email · CSV
Chunk	LangChain splitters · Stable IDs
Index	sentence-transformers · Chroma · FAISS
Search	rank_bm25 · Hybrid fusion · Reranker
Answer	LangGraph · Groq / Gemini / Ollama
Serve	FastAPI · pgvector · Docker

Plain Python first for every mechanism, then the library.

What it must get right

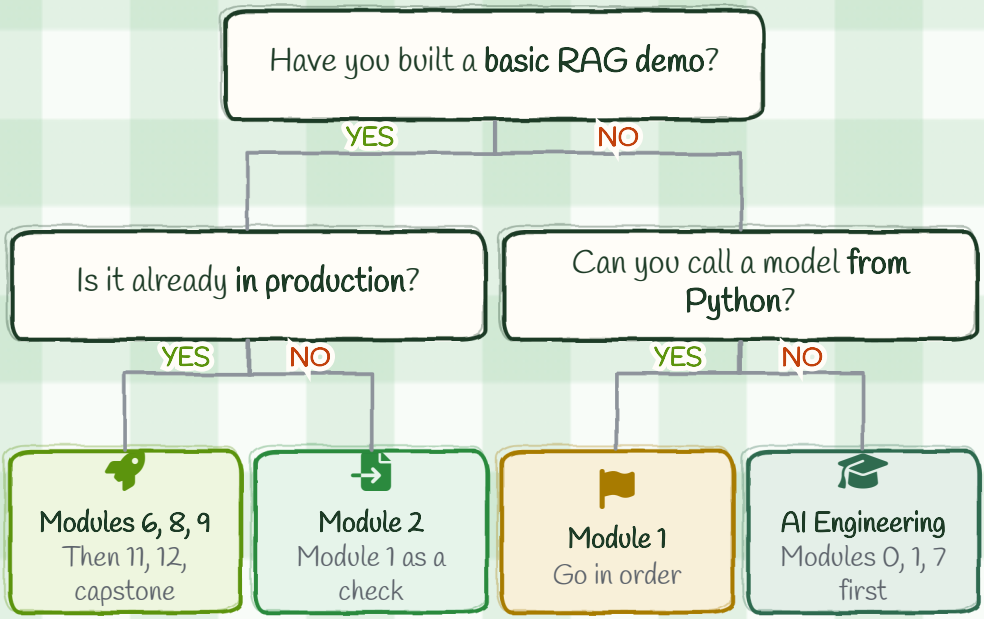
Six Rules, Mapped to Modules

THE ASSISTANT MUST	WHERE YOU LEARN IT
Cite a source for every claim	Module 9
Say "I don't know" and hand over	Modules 9, 13
Never show a doc they can't see	Modules 5, 12
Prefer current over archived	Module 6
Treat document text as data	Module 9
Prove it, and keep proving it	Module 11

Live facts like a repair status? **Query the database**, not the docs.

Quick check

Where Should You Start?



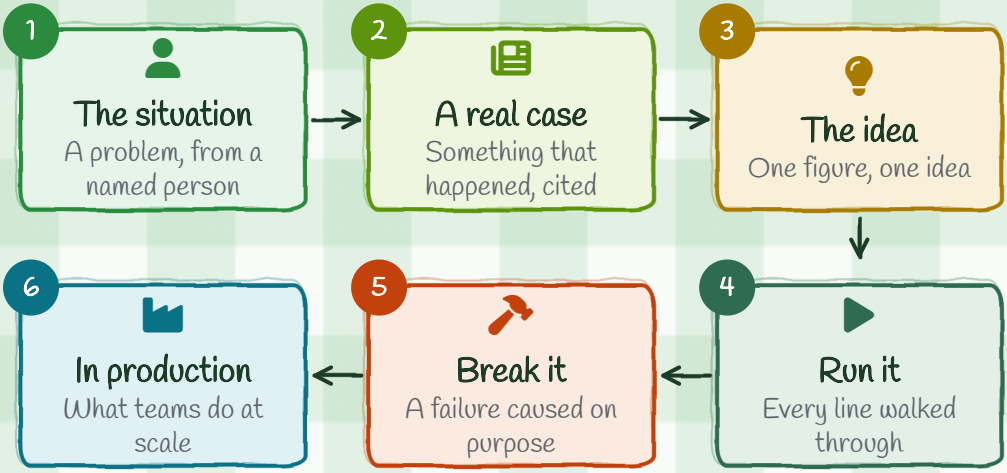
Every module lists what you need first, so you **skip safely.**

How each lesson runs

Learn It by Breaking It

One example, start to finish

Every lesson walks the same path, then ends with a self-check.



Each module ends with a **lab** and a **real incident**, replayed and fixed.

WE BUILT THIS ROADMAP INTO A COURSE

Build RAG That Cites and Holds Up

All 14 steps, on one assistant you build piece by piece

- 14 modules, from first answer to production service
- One running project: a support assistant over messy company docs
- Runs in Google Colab with a free Groq or Gemini key
- Plus the RAG Interview Kit: 405 answered questions

Start the RAG course ↓

practicai.tech

Questions? Comment below and I'll answer every one.